Java Skill Matrix

Skill	Must have	Nice to have	Topics
Java Web Development	V		HTTP, sockets, URLConnection, Servlet containers and Application Servers (Tomcat, Jboss, etc.), JSP, Servlet, Spring MVC
Java Observability	V		Logging, Log4J, Appenders, SLF4J, Tracing
Java Dependency Injection	V		IoC, DI, Spring Core, AOP
Java Microservice Infrastructure		V	Microservices architecture, Spring Cloud, Design Patterns
Java Bootstrapping		V	Spring boot
Java Security		V	Authentication, JWT Token, Spring Security
Java Persistence		V	ORM, Hibernate, JPA, Spring Data,
Java SQL	V		RDBMS concepts, SQL, JDBC, JTA, ACID
Java NoSQL		V	NoSQL overview, NoSQL DB types, Key-Value databases, Document databases, Column family stores, Graph Databases, CAP theorem
Java Messaging		V	Event Driven Architecture, Kafka, ActiveMQ, PubSub
Java Build	V		Maven
Java Version Control		V	Git
Java Troubleshooting	V		Error handling, Debugging, Monitoring, Debugging of JVM
Web Services		V	XML, XSLT, XPATH, JSON (GSON-Jackson), REST, SOAP & WSDL, Spring REST
AI		V	Chat GPT, Copilot GitHub

Testing		V	TDD, Test Pyramid, Junit, TestNG, Mocking, Test coverage
Software Design		V	Design patterns (GoF and GRASP), SOLID, UML, Code Convention, Static code analyzing (SonarLint, Checkstyle, Spotbugs), DAL, DAO, Client - Server, MVC
Software Engineering Practices		V	CI/CD, Code Qualit, Branching strategies
Software Engineering Processes		V	Estimation SDLC Agile Scrum Kanban
Teamwork and Collaboration		V	
Networking	V		TCP, HTTP
Software Engineering Knowledge & Experience		V	Deployment Strategies, Development Experience, Cryptography, Operating Systems
HTML	V		A thorough understanding of HTML - mastery over using various tags, attributes, and elements to construct a semantically correct, accessible, and SEO-optimized website structure. (HTML standards, Accessibility, SEO basics, Forms, Validations)
CSS	V		CSS requires proficiency in applying styles, layouts, and animations to enhance web page aesthetics and user experience while ensuring responsiveness across different devices. Making Layouts, Positioning, Display, Box Models, CSS Grid Flexbox Responsive Design, and Media Queries, Familiarity with CSS preprocessors, such as Sass or Less, and how to use them to enhance CSS.
JavaScript	V		Syntax and Basic Constructs, ES6+, and modular JavaScript. Understand the concepts, Hoisting, Event Bubbling, Scope, Prototype, Shadow DOM, strict
JavaScript in Browser	V		DOM Manipulation, XMLHttpRequest (XHR) and Fetch API object to make HTTP requests from your JavaScript code
Web Communication Protocols	V		HTTP, URL, HTTPS, FTP, DNS, SMTP, POP3 and IMAP, WebSocket, REST, SOAP
JavaScript Development Tools	V		JavaScript Build Tools, Basic Development Tools, Browser Developer Tools, Task runners, Bundlers, Transpilers, Linting, Hinting, Microfrontent building tools, Monorepos
Browser APIs	V		Manipulate cookies in the browser using JavaScript, Local Storage API, Web Storage API, Geolocation API, Web Workers, Service Workers
TypeScript		V	TypeScript code using its constructs: Classes, Interfaces and types, Unions, Functions types, Understanding of module system, syntax, resolution
JavaScript Top Frameworks		V	React or Angular | Vue (HW: TodoList)
JavaScript Cross-Mobile Platform		V	JavaScript Mobile Frameworks or libraries such as React Native, PhoneGap, and Ionic.
JavaScript Desktop Platform		V	Building desktop applications with JavaScript - Electron.js, Proton Native
Web Performance Analysis and Optimization		V	Project Framework Optimization Techniques, Page load optimization techniques, Repaint/Reflow Understanding
Node.js Basics		V	Basic commands, Build-in Node.js CLI options, Node.js API, Node.js Deployment - NPM Scripts

Cloud computing concepts		\/	Cloud Computing definition and characteristics Service models (SaaS, PaaS, IaaS) Deployment models (Private, Community, Public, Hybrid cloud)
Cloud services (AWS)		\/	Compute Services (EC2, Elastic Beanstalk, Auto Scaling, Lambda, Batch, Serverless Application Model (SAM)) Storage (S3, Glacier, EBS, Amazon EFS, AWS Backup, AWS Datasync, AWS Transfer Family) Database (RDS, ElastiCache, DynamoDB, Redshift, Aurora) Networking and Content Delivery (VPC, Route53, Elastic Load Balancing, CloudFront, API Gateway, Transit Gateway, WAF, Direct Connect) Identity (IAM, AWS Organizatrions, AWS Control Tower) Data and Infrastructure Protection (Secrets Manager, Certificate Manager, Web Application Firewall, Key Management Service) Containers (ECS, Fargate, ECR, EKS) Developer Tools (Amplify, Cloud9, CloudShell, CodeArtifact, CodeBuild, CodeCommit, CodeDeploy, CodeGuru, CodePipeline, CodeStar, , X-Ray) Analytics (Athena, Kinesis, OpenSearch Service) Application Integration (AppSync, EventBridge, SNS, SQS, Step Functions) Management and Governance (AppConfig, CDK, CloudFormation, CloudWatch, CloudTrail, CLI)
IaC Concepts		\/	Overview and advantages Approaches (declarative and imperative, examples) Methods (pull, push)
Infrastructure as Code		\/	Terraform
Containerization		\/	Docker Kubernetes
Observability and troubleshooting		\/	Prometheus, Grafana